

Plasma homocysteine as a risk factor for dementia and Alzheimer's disease.

BACKGROUND: In cross-sectional studies, elevated plasma homocysteine levels have been associated with poor cognition and dementia. Studies of newly diagnosed dementia are required in order to establish whether the elevated homocysteine levels precede the onset of dementia or result from dementia-related nutritional and vitamin deficiencies. **METHODS:** A total of 1092 subjects without dementia (667 women and 425 men; mean age, 76 years) from the Framingham Study constituted our study sample. We examined the relation of the plasma total homocysteine level measured at base line and that measured eight years earlier to the risk of newly diagnosed dementia on follow-up. We used multivariable proportional-hazards regression to adjust for age, sex, apolipoprotein E genotype, vascular risk factors other than homocysteine, and plasma levels of folate and vitamins B12 and B6. **RESULTS:** Over a median follow-up period of eight years, dementia developed in 111 subjects, including 83 given a diagnosis of Alzheimer's disease. The multivariable-adjusted relative risk of dementia was 1.4 (95 percent confidence interval, 1.1 to 1.9) for each increase of 1 SD in the log-transformed homocysteine value either at base line or eight years earlier. The relative risk of Alzheimer's disease was 1.8 (95 percent confidence interval, 1.3 to 2.5) per increase of 1 SD at base line and 1.6 (95 percent confidence interval, 1.2 to 2.1) per increase of 1 SD eight years before base line. With a plasma homocysteine level greater than 14 micromol per liter, the risk of Alzheimer's disease nearly doubled. **CONCLUSIONS:** An increased plasma homocysteine level is a strong, independent risk factor for the development of dementia and Alzheimer's disease.